Sahaj Bhadja

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Summary

AI-native software engineer with 2+ years building production LLM workflows, full-stack products, REST APIs, and testing automation. Experienced in TypeScript, React, Next.js, Node.js, Python, PostgreSQL, code reviews, documentation, and reliable delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Reduced user-facing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling Python API request paths and tuning Azure SQL queries, indexes, and secure access patterns.
- Built FastAPI services and fault-tolerant asynchronous LLM workflows for 5,000+ monthly jobs, using explicit states, retries, reconciliation, and failure isolation to triple throughput while sustaining 99%+ reliable production processing.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask APIs and React implementation workflows supporting 150+ users and 10,000+ daily requests, integrating secure ingestion, authentication, AWS RDS persistence, status tracking, and instructor-facing product experiences.
- Designed concurrent processing for 1,000+ uploaded files with real-time job-state updates, automated validation, and fault-tolerant execution, reducing turnaround 60% while maintaining stable API performance during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying structured automated tests, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a performance-critical architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Maintained 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving confidence in cross-platform production releases.
- Partnered with Microsoft and GitHub stakeholders on API integrations, then optimized filtering, pagination, indexing, and caching to improve client response times 30% and production data-pipeline efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js services with WebSocket synchronization across eight customer, billing, scheduling, permissions, and notification workflows, supporting 2 million+ Azure SQL records through maintainable service boundaries.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure services, reducing release time 50% while improving software quality, feedback speed, and operational reliability.

Skills

Languages & Product: TypeScript, JavaScript, Python, Java, C++, React, Next.js, Node.js, Flask, FastAPI, REST APIs, responsive interfaces, full-stack development
AI & Agent Systems: LLM orchestration, multimodal LLMs, semantic search, vector retrieval, FAISS, LLaMA-2, asynchronous workflows, evaluation, validation, failure isolation, prompt workflows
Data & Cloud: PostgreSQL, AWS RDS, Azure SQL, MySQL, MongoDB, AWS, Azure, GCP, S3, Blob Storage, indexing, caching, latency tuning, data modeling
Testing & Delivery: GitHub, Git, CI/CD, Jenkins, GitHub Actions, Docker, PyTest, Mockito, unit testing, integration testing, regression testing, code review, debugging, observability, documentation, Agile, production support

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-assisted SQL developer workspace supporting 10+ databases, combining Java and TypeScript services, schema-aware query generation, validation, Docker, and cross-platform workflows for reliable data exploration and management.

PiXMod | Live | *Next.js, TypeScript, MongoDB, CLIP, Transformers*
- Built an AI-powered SaaS product with Next.js, TypeScript, CLIP, SAM, and Stable Diffusion, processing 1,000+ edited assets through secure subscriptions, model integrations, and MongoDB-backed workflows at scale.

Big-Brain | Live | *Python, PostgreSQL, Clerk, GCP, REST APIs*
- Built semantic retrieval over 50,000+ documents using LLaMA-2, FAISS, PostgreSQL, and GCP, enabling contextual question answering and summarization with optimized pipelines that reduced retrieval latency 40%.